QMIM: HW 1 – Portfolio Theory 1

Theoretical Exercises:

1) As shown yesterday in class, solve the minimization problem: min ($w'\Sigma w$) subject to $w'R = r_p$ and $w'\mathbf{1} = 1$ to obtain the portfolio weights that minimize variance for a given level of expected return. Then, find the minimum variance portfolio over all levels of expected return.(4 points)

2) Can short interest be greater than 100%? Why? Why is it important to know what the short interest of an asset is before shorting it?(2 points)

Practical Exercises:
It is strongly recommended (read: compulsory) to use Python as it is the standard coding language for data analysis in the industry!

1) Simulating and Printing the Investment Set and the Efficient Frontier(4 points):

- From Yahoo Finance, download the daily adj close prices for GOOGL, AMZN, MSFT and AAPL from 01-01-2021 to 12-31-2021.
- Compute log returns and build 1000 different portfolios using random weighting for these 4 stocks). Suggestion: use np.random.seed(<any int>) so that you have the same 1000 weights every time you run the notebook.
- Compute expected return, volatility and Sharpe ratio for each portfolio. Remember to annualize returns and volatilities!
- Investment Set: Plot the data (scatter plot, x axis = volatilities, y axis = returns) and highlight with a different color the portfolio with the highest SR.
- Efficient Frontier: Plot it against the investment set.
- What portfolio would you chose? What considerations do you need to make in order to take this decision?
- How would investors' personal needs impact the choice of their portfolio?
- Find the minimum variance portfolio by:
    - Using the formula derived in exercise 1
    - Ranking the simulated portfolios
    - By finding the set of weights s.t. the covariance of each asset to the portfolio is equal (please refer to what explained in class)

Theoretical Exercises:

1) As shown yesterday in class, solve the minimization problem: min (w'Σw) subject to w'R = r_p and w'**1** = 1 to obtain the portfolio weights that minimize variance for a given level of expected return. Then, find the minimum variance portfolio over all levels of expected return.(4 points)

$\min(w'\Sigma w)$ $\qquad$ $w'R = r_p$, $\quad w'\underline{1} = \underline{1}$

$$\tilde{r}_p = \sum_{j=1}^{N} w_j \tilde{r}_j = w'r$$

$$E[\tilde{r}_p] = \sum_{j=1}^{N} w_j E[\tilde{r}_j] = w'\mu$$

As $w'\mu = \underline{1} \Rightarrow \mu = \underline{1}$.

$$\sigma^2(r_p) = w^2\sigma^2(r_A) + (1-w)^2\sigma^2(r_B) + 2w(1-w)\mathrm{cov}(r_A, r_B)$$

* For a given expected Return :

$$L(x, \lambda) = f(x) - \lambda \cdot g(x)$$

$$L(x) = \frac{1}{2} w' \Sigma w - \lambda_1 (w'R - r_p) - \lambda_2 (w'1 - 1)$$

$$\frac{\partial L}{\partial w} = \Sigma w - \lambda_1 R - \lambda_2 1 = 0$$

$$\frac{\partial L}{\partial \lambda_1} = -(w'R - r_p) = 0$$

$$\frac{\partial L}{\partial \lambda_2} = -(w'1 - 1) = 0$$

$$w = \lambda_1 \Sigma^{-1} R + \lambda_2 \Sigma^{-1} 1.$$

Substituted w into the two constraint equations:

$$R' (\lambda_1 \Sigma^{-1} R + \lambda_2 \Sigma^{-1} 1) = r_p \quad 1' (\lambda_1 \Sigma^{-1} R + \lambda_2 \Sigma^{-1} 1) = 1$$

Let $A = R' \Sigma^{-1} R$, $B = R' \Sigma^{-1} 1$, $C = 1' \Sigma^{-1} 1$.

$$\Rightarrow \lambda_1 A + \lambda_2 B = r_p \quad \lambda_1 B + \lambda_2 C = 1.$$

$$\Rightarrow \lambda_1 = \frac{C r_p - B}{AC - B^2}, \quad \lambda_2 = \frac{A - B r_p}{AC - B^2}$$

$$\Rightarrow w = \frac{C r_p - B}{AC - B^2} \Sigma^{-1} R + \frac{A - B r_p}{AC - B^2} \Sigma^{-1} 1.$$

* For variance :

$$L = \frac{1}{2} w' \Sigma w - \lambda (w' 1 - 1)$$

$$\frac{\partial L}{\partial w} = \Sigma w - \lambda 1 = 0$$

$$\frac{\partial L}{\partial \lambda} = -(w' 1 - 1) = 0$$

As $w = \lambda \Sigma^{-1} 1$.

$$\Rightarrow 1' (\lambda \Sigma^{-1} 1) = 1$$

$$\lambda (1' \Sigma^{-1} 1) = 1$$

$$\lambda C = 1 \quad \lambda = \frac{1}{C}$$

$$\Rightarrow w = \frac{1}{C} \Sigma^{-1} 1.$$

$\Rightarrow$ Minimum variance $\Rightarrow w = \frac{1}{C} \Sigma^{-1} 1.$

#2.

Yes, short interest can be greater than 100% because shares can be borrowed, sold, repurchased many times.

We need to check short interest because high short interest increase the risk of a short and borrowing cost.